



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

4.0 IIoT Reference Architecture

July 2025

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Records of Change

Date of Issue/ Reissue	Revision No.	Reason of Development / Change
July 2025	00	New module offering inline with TI&R 2025.

Course Content

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Learning Outcome

At the end of the session, you will be able to:

- Understand the five-tier topology of IIoT and how each layer contributes to a seamless, efficient industrial ecosystem.
- Learn how edge computing enables real-time processing, reducing latency and improving industrial automation.
- Explore the role of networking in IIoT for industrial applications.
- Identify the significance of platforms in managing, analyzing, and processing IoT-generated data.

04. IIoT Reference Architecture

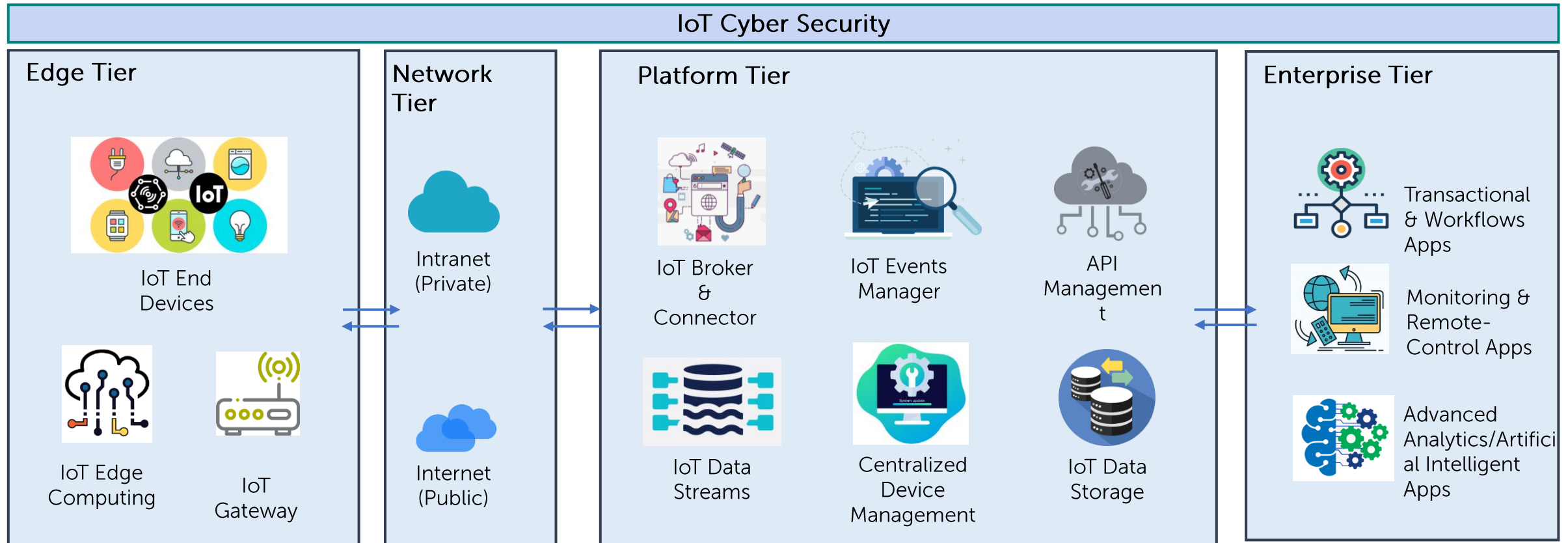
- Overview of IIoT Reference Architecture (5-Tier Topology)
- Edge Tier
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Overview of IIoT Reference Architecture

IIoT Architecture is built upon 4 main Tiers



Edge tier is where data is sampled and collected from the environment by instrumented "things" or devices.

Network tier is where data is transmitted between Edge & Platform Tier

Platform tier is where the IIoT system aggregates systemwide data and events from many edge locations.

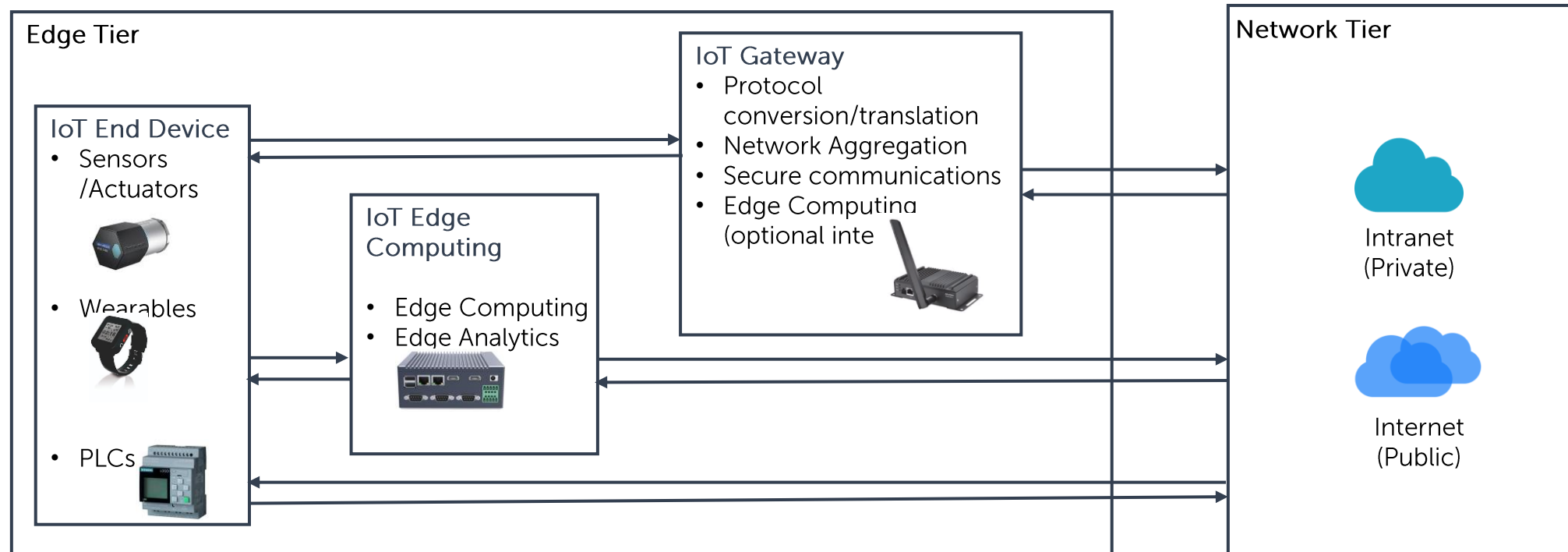
Enterprise tier is where IIoT integrates with the set of applications, processes and services required to accomplish a business objective.

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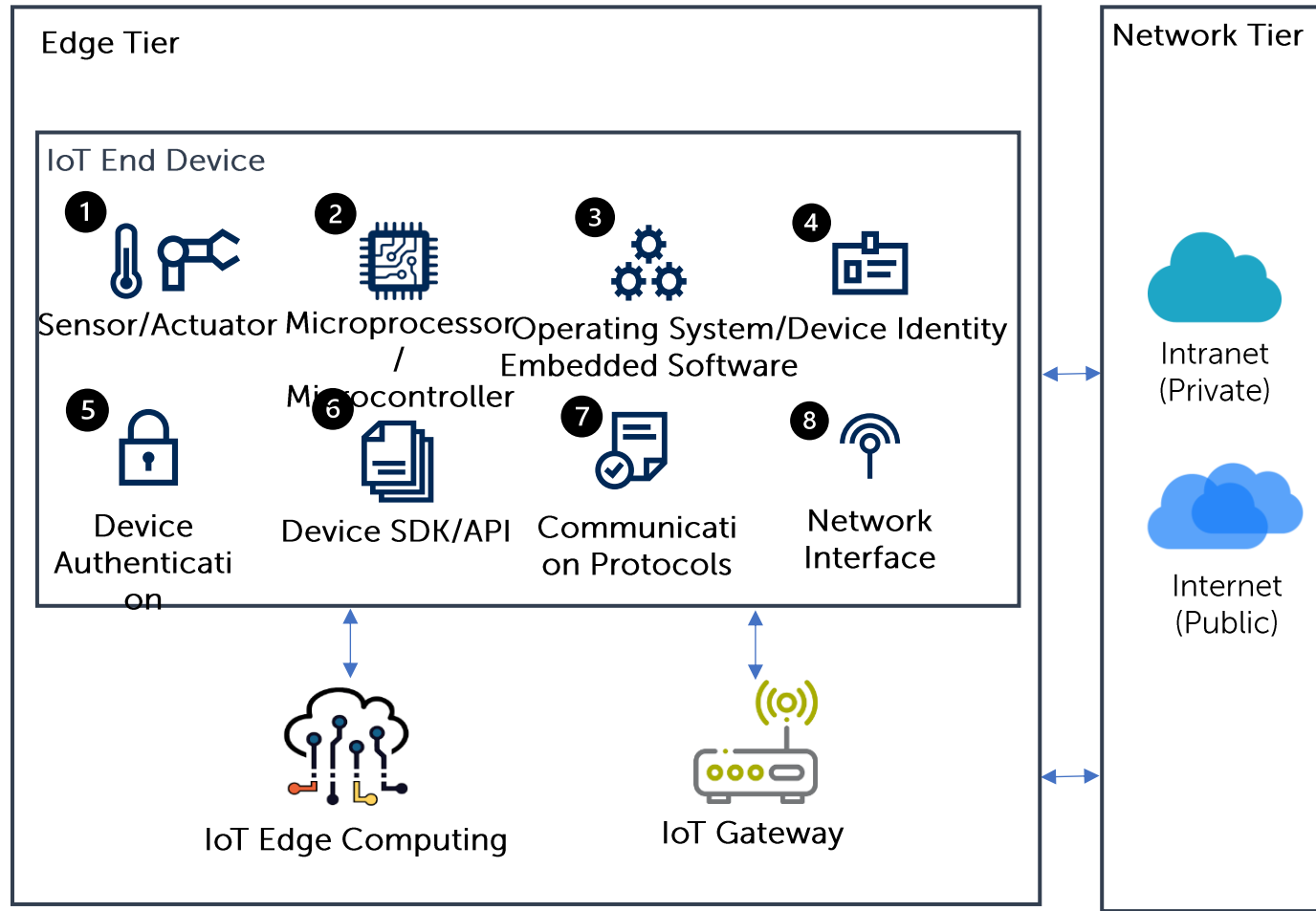
Edge Tier

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Edge Tier

IoT End Devices



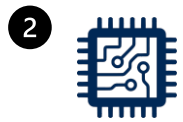
Edge Tier

IoT End Devices



Sensor/Actuator

- a. The sensors/actuators on IoT End Devices is the main component that interacts with the physical world and will be the main requirement on the determining the characteristics and applicability of the IoT End Devices.
- b. Predominantly, most of the use cases for IoT will only require the use of sensors, however there will be cases where actuators will be required for controlling the physical environment.
- c. Use cases will further determine the type of sensors required e.g. Temperature, Strain, Pressures and etc



Microprocessor
/
Microcontroller

- a. Depending on the type of IoT End Device, the main processing units can be categorized into the 2 following broad category:
 - i. Microprocessor :
 - will have a dedicated CPU with higher processing power, normally based on x86-64 or ARM architecture and will generally run a general-purpose OS, such as Windows, Linux
 - ii. Microcontroller:
 - will contain many of the functions, such as memory, interfaces, and I/O within the chip itself and will often use an embedded OS/software



Operating System/
Embedded Software

- a. Operating System (OS) in IoT devices provides the foundational software layer that manages hardware resources, enables device functionality, and supports communication.
- b. The operating system (OS) and Embedded Software onboard the IoT End Devices shall be categorized into the 3 main categories
 - i. General OS:
 - Is based on popular OS such as Android, Windows, Linux and others.
 - ii. Embedded OS:
 - OS designed specifically to work with specific function and within the strict constraint of the IoT End Devices. Example of Embedded OS are Free RTOS,
 - iii. Embedded Software:



Device Identity



- a. IoT Device Identity refers to a unique identifier and associated credentials assigned to an IoT device to distinguish it from others within a network or ecosystem. This identity is critical for managing, securing, and interacting with IoT devices effectively.
- b. The type of identifier will usually depend on the category of the IoT End Device and the Network/Communication Protocols
- c. Consequently, these unique identifiers will be required for registration and management of the device within each tier of the IoT building blocks

Edge Tier

IoT End Devices



5 Device Authentication

- a. IoT Device Authentication ensures that IoT devices are securely verified before they connect to a network or communicate with other devices, services, or the cloud. It is a crucial aspect of securing IoT systems against unauthorized access, data breaches, and device impersonation.
- b. The method employed for device authentication will be dependent on the system that the IoT End Device will be connected to within the following IoT tier:



6 Device SDK/API

- i. Network Tier: For example, WPA2-PSK is typically being used for IoT device to authenticate to WiFi Networks
- ii. Platform Tier: For example, X.509 certificate-based authentication is typically being used to authenticate to an IoT broker
- a. An IoT device Software Development Kit (SDK), is collection of tools, libraries, and documentation that helps developers create software for IoT devices or applications. SDK provides pre-built functions for tasks like data collection, device management, connectivity, and security.
- b. An IoT Application Programming Interface (API) are asset of defined methods and protocols that enable software applications to communicate with IoT devices, platforms, or services.
- c. SDK and API ensures capability of the IoT device can be fully leveraged when integrating the devices in an IoT systems



7 Communication Protocols

- a. IoT communication protocols are communication standards that enable IoT devices to connect, exchange data, and interact with other devices, applications, and platforms.
- b. IoT communication protocols standards ensures that the IoT devices are interoperable with other components within an IoT systems
- c. Some of the popular IoT protocols that are being used includes MQTT and OPC UA

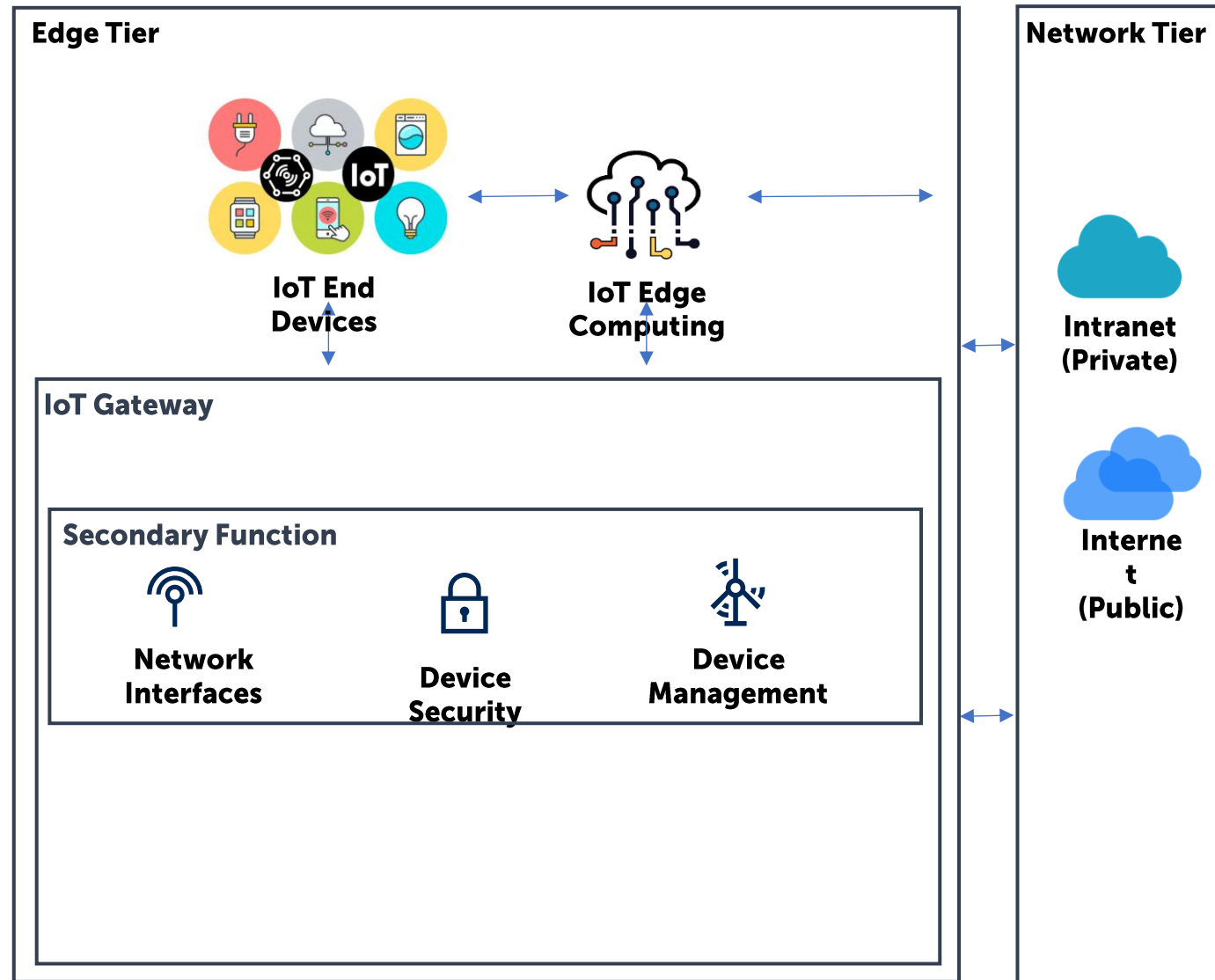


8 Network Interface

- a. The network interfaces on the IoT device facilitates the data across networks
- b. Commonly used network interfaces for IoT devices can be wired such as Ethernet or wireless such as 5G or WiFi
- c. There are also specialized wireless interfaces optimized for IoT such as BLE and LoRaWAN

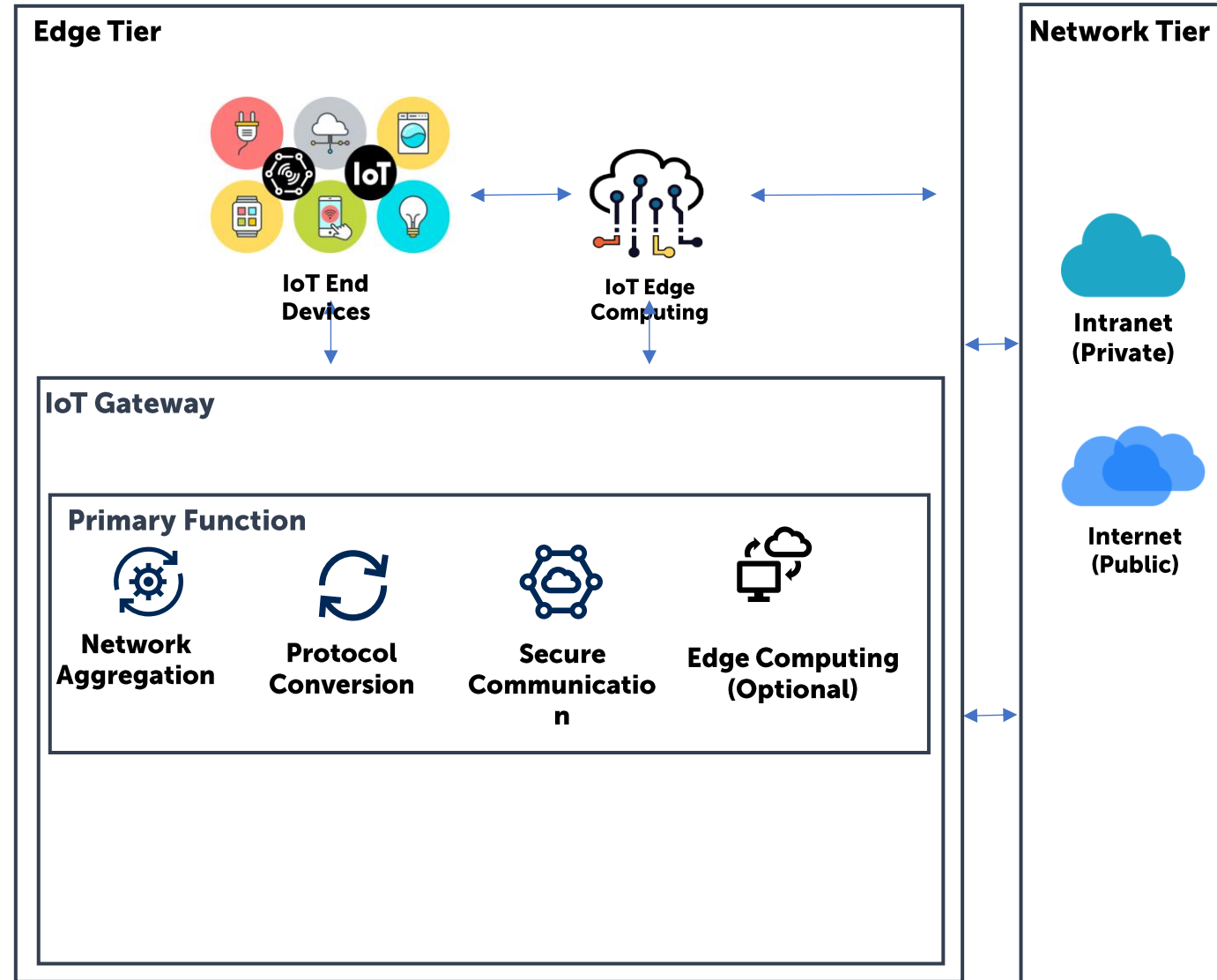
Edge Tier

IoT End Devices



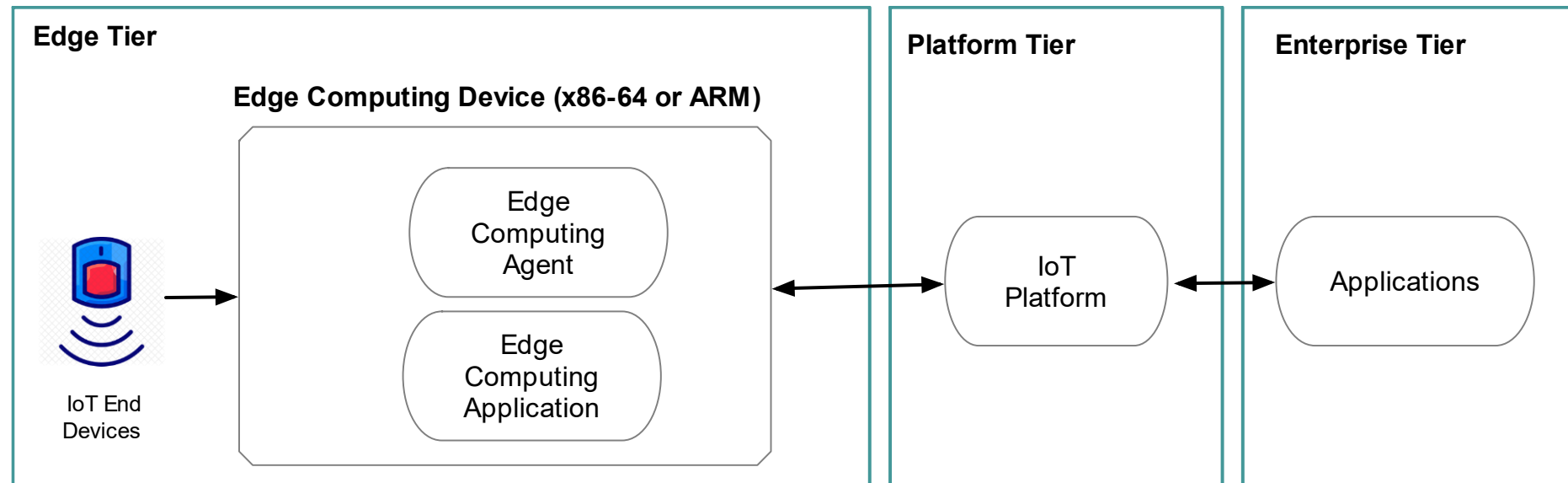
Edge Tier

IoT Gateway



Edge Tier

Edge Computing

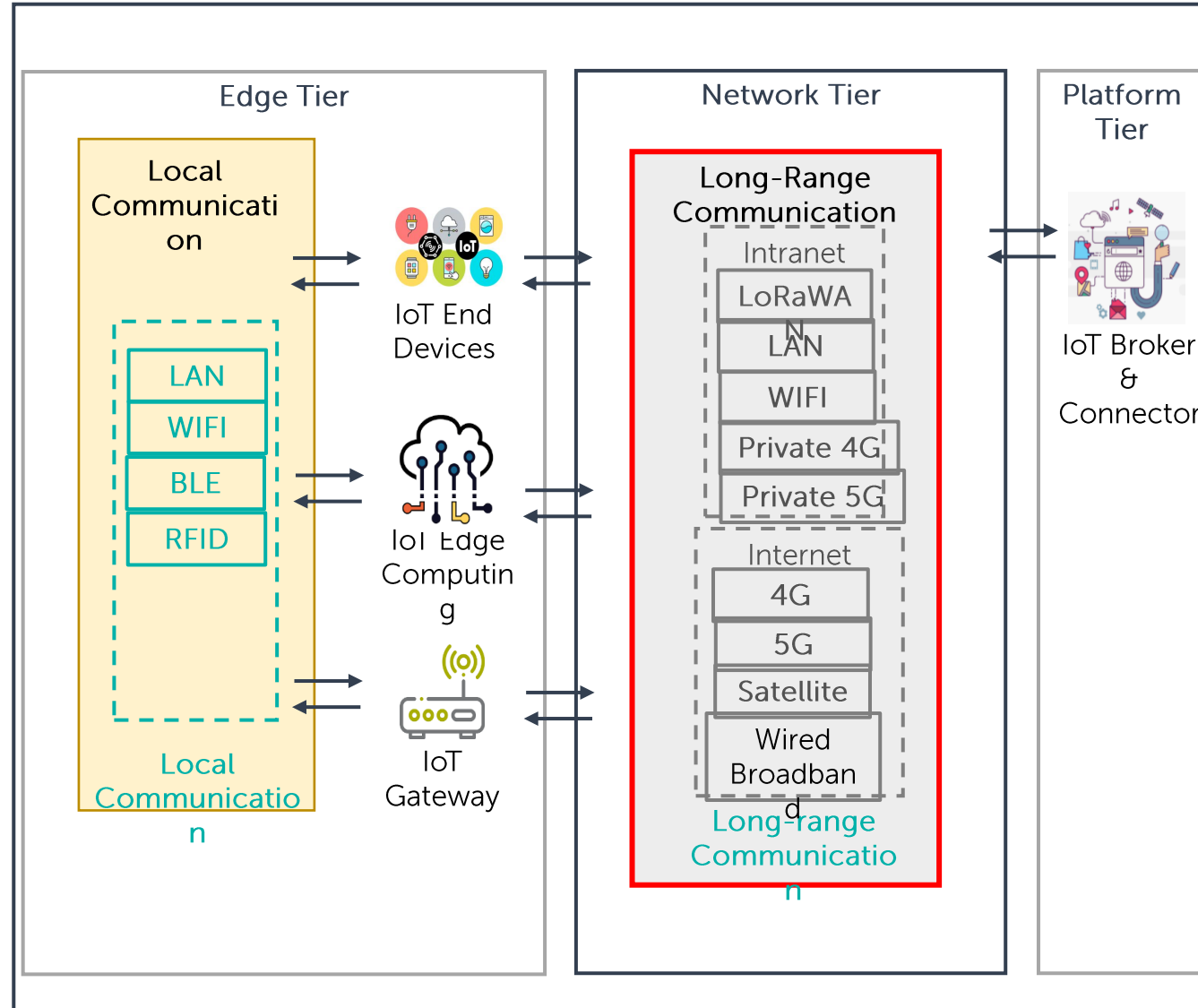


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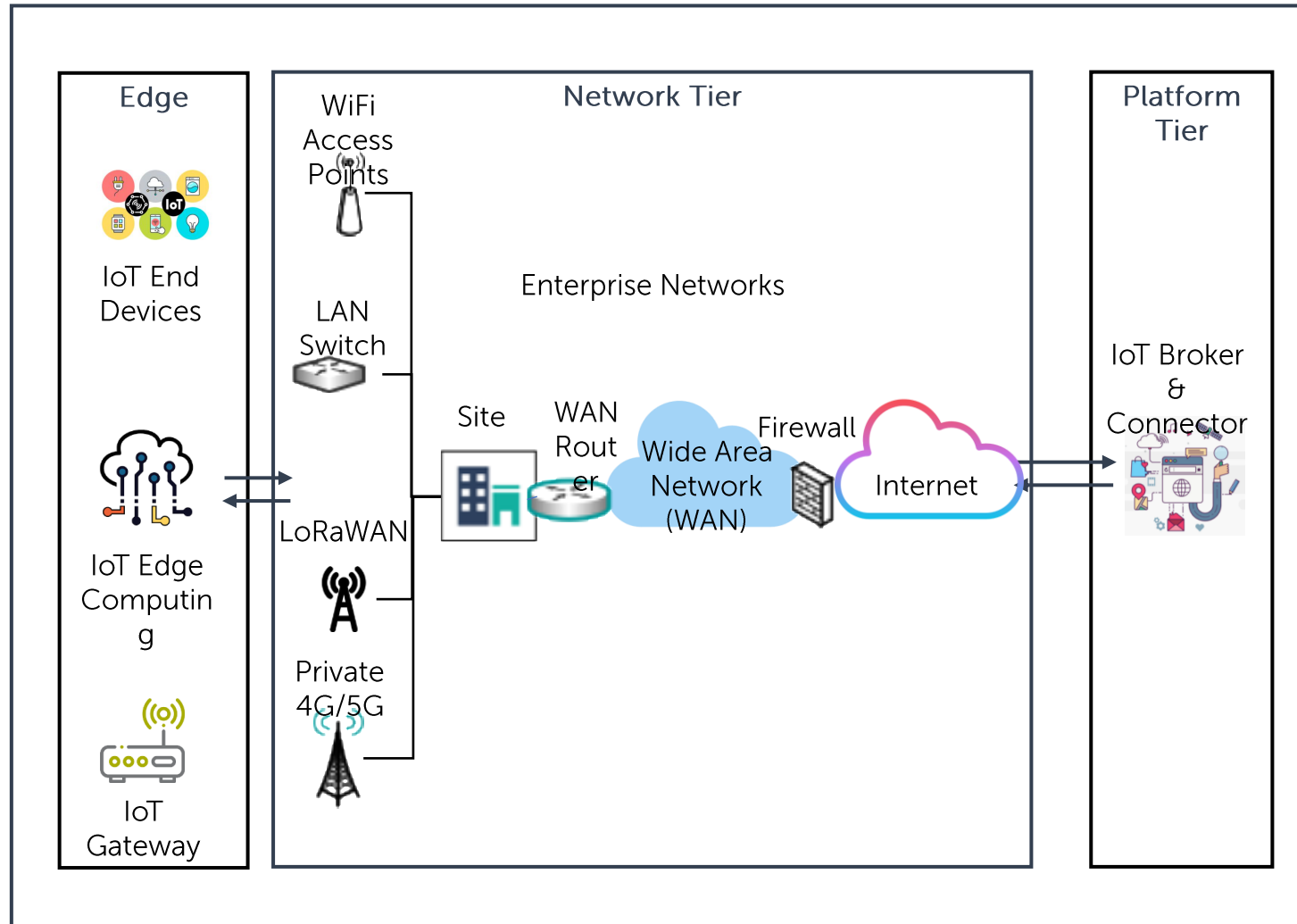
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Network Tier

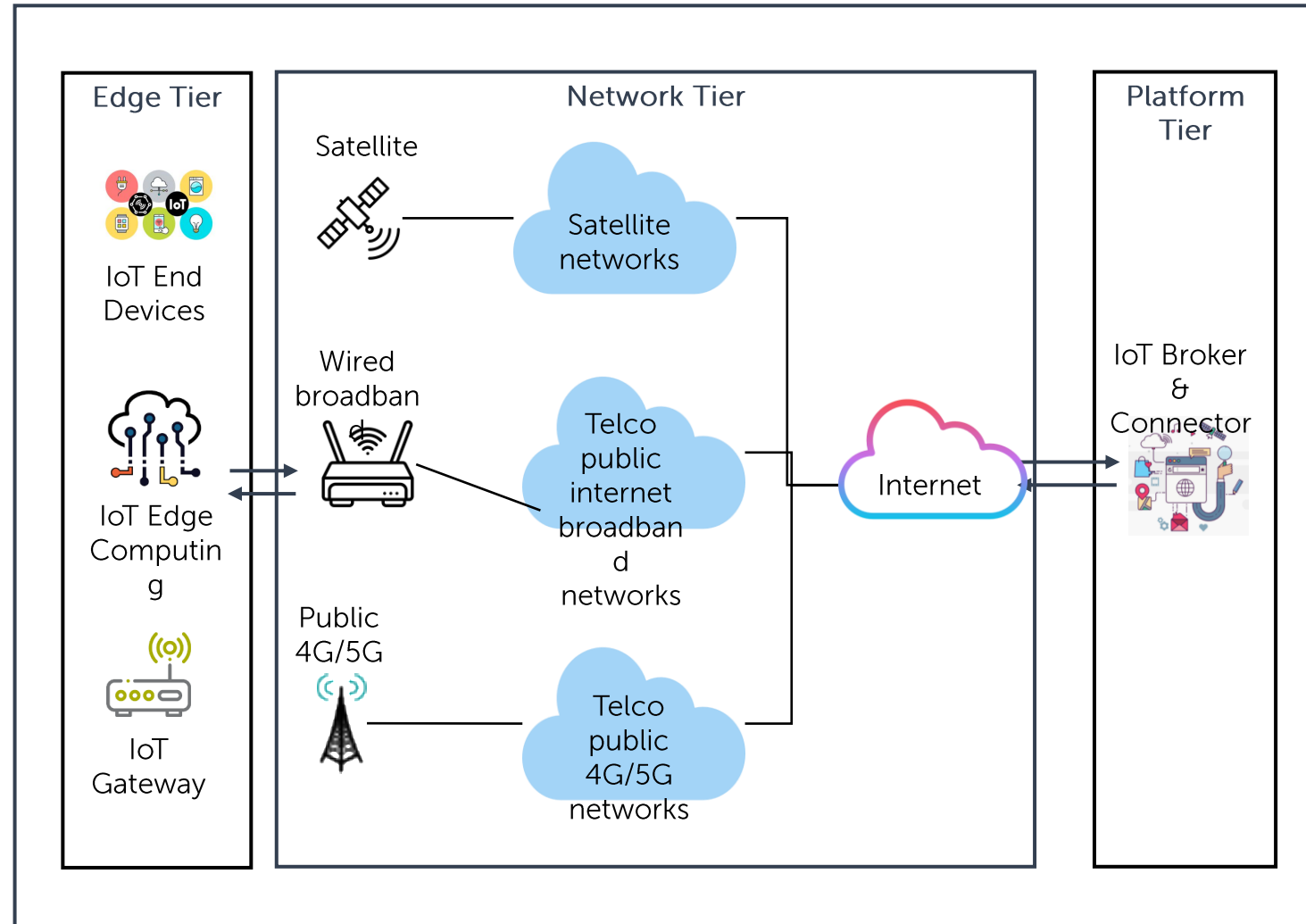
Network Tier is where data is transmitted between Edge Tier and Platform Tier



Network Tier – Intranet (Private)



Network Tier – Internet (Public)

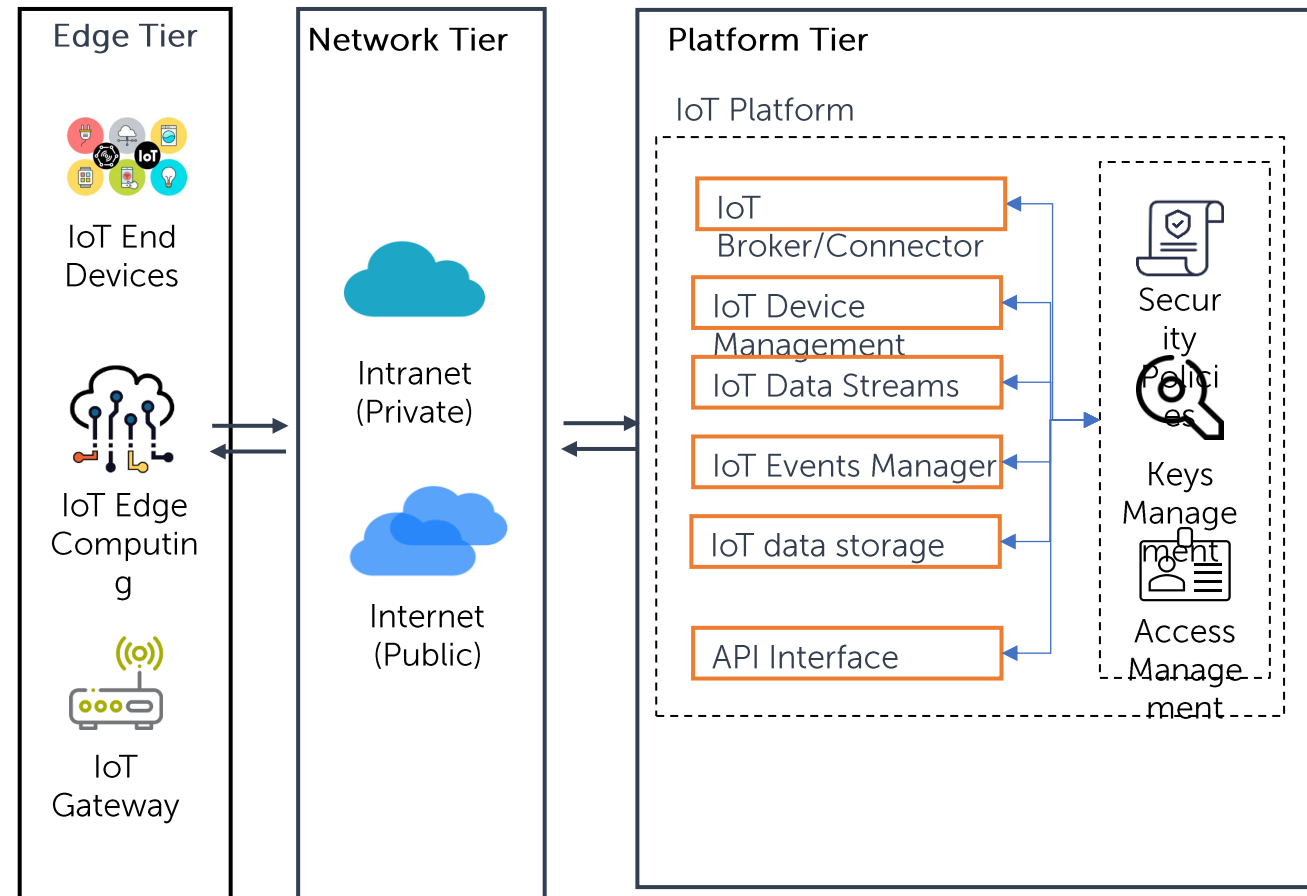


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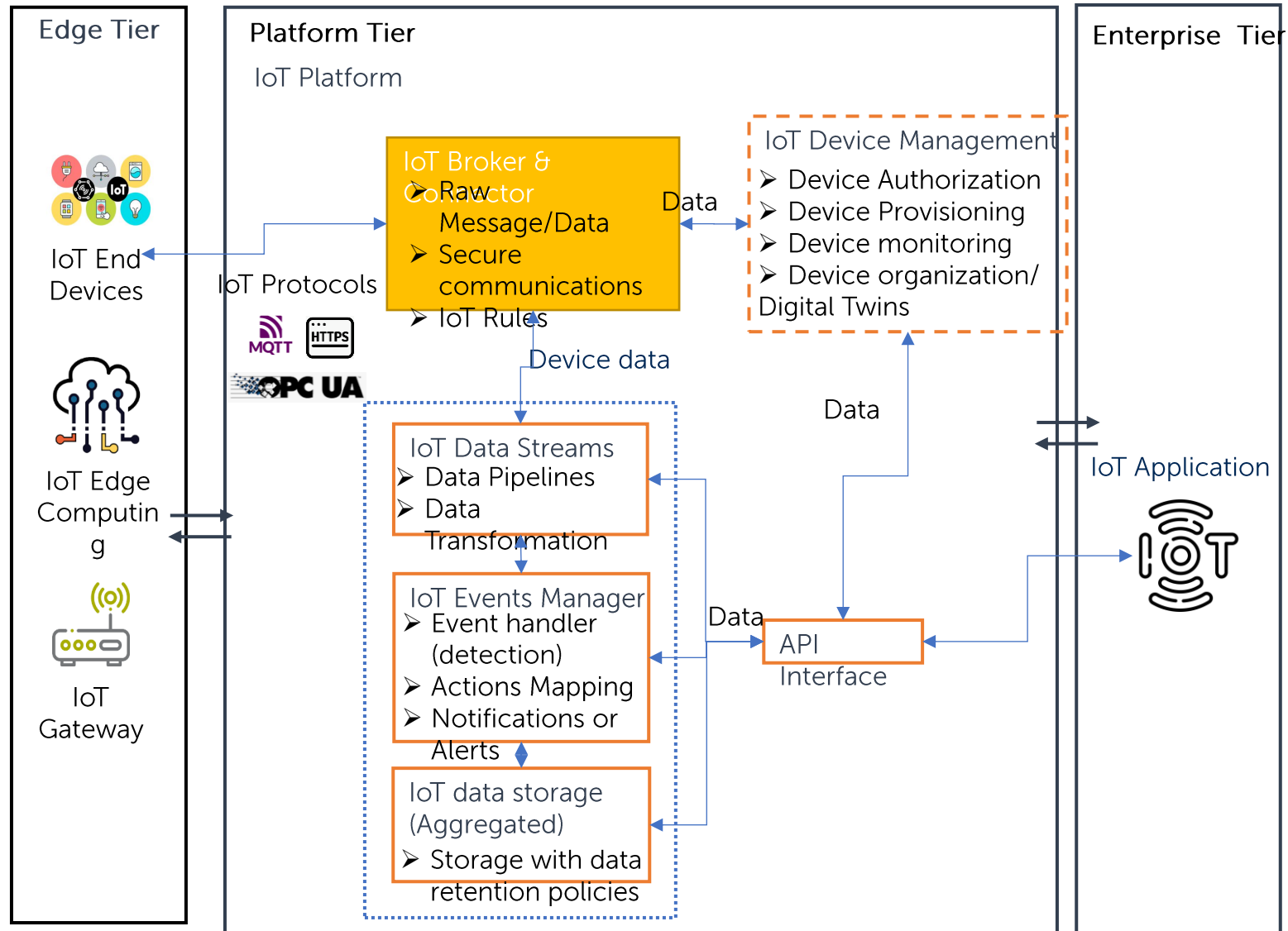
Platform Tier

Platform tier is where the IoT system aggregates systemwide data and events from many edge locations.



Platform Tier

IoT Broker/Connector

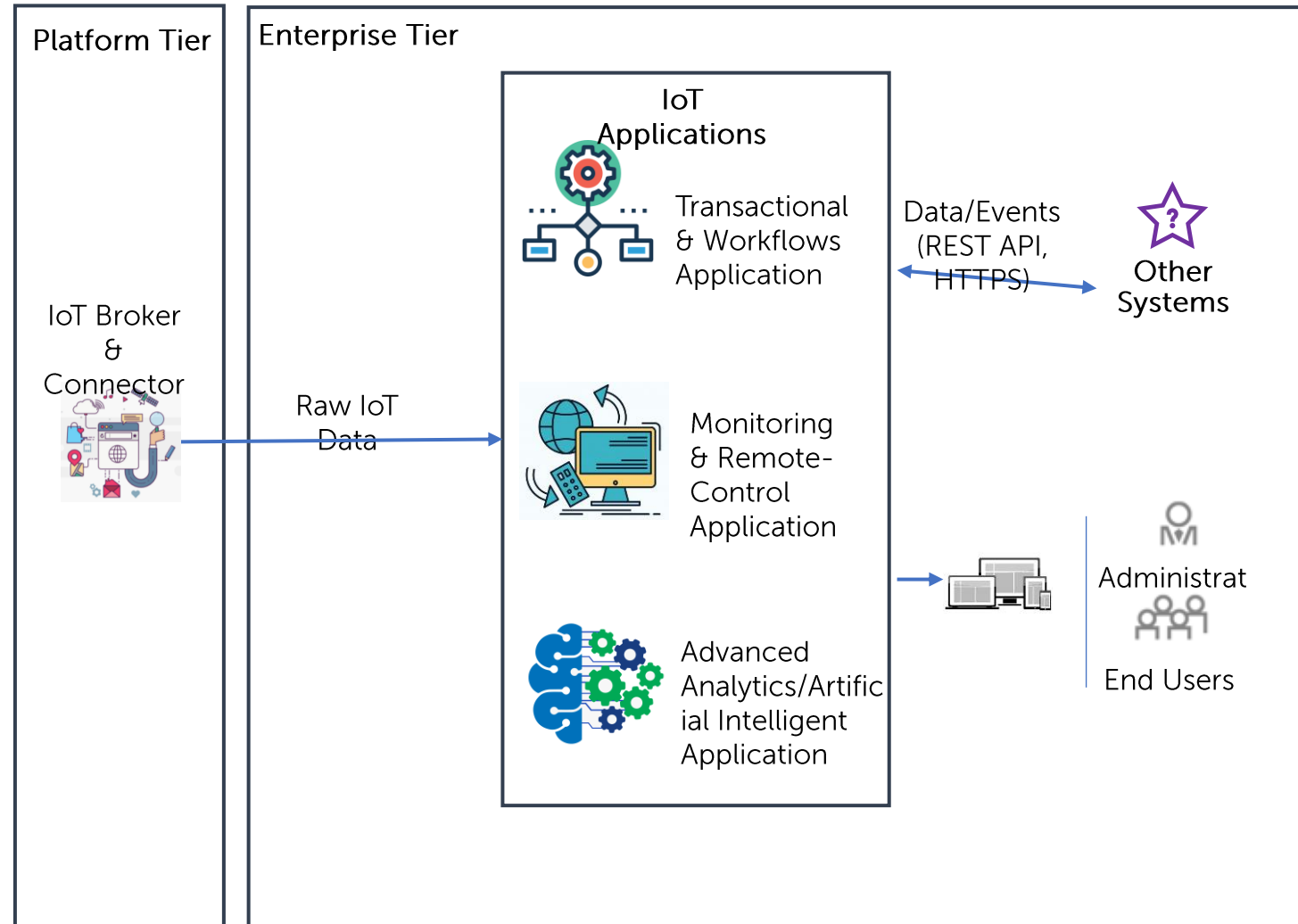


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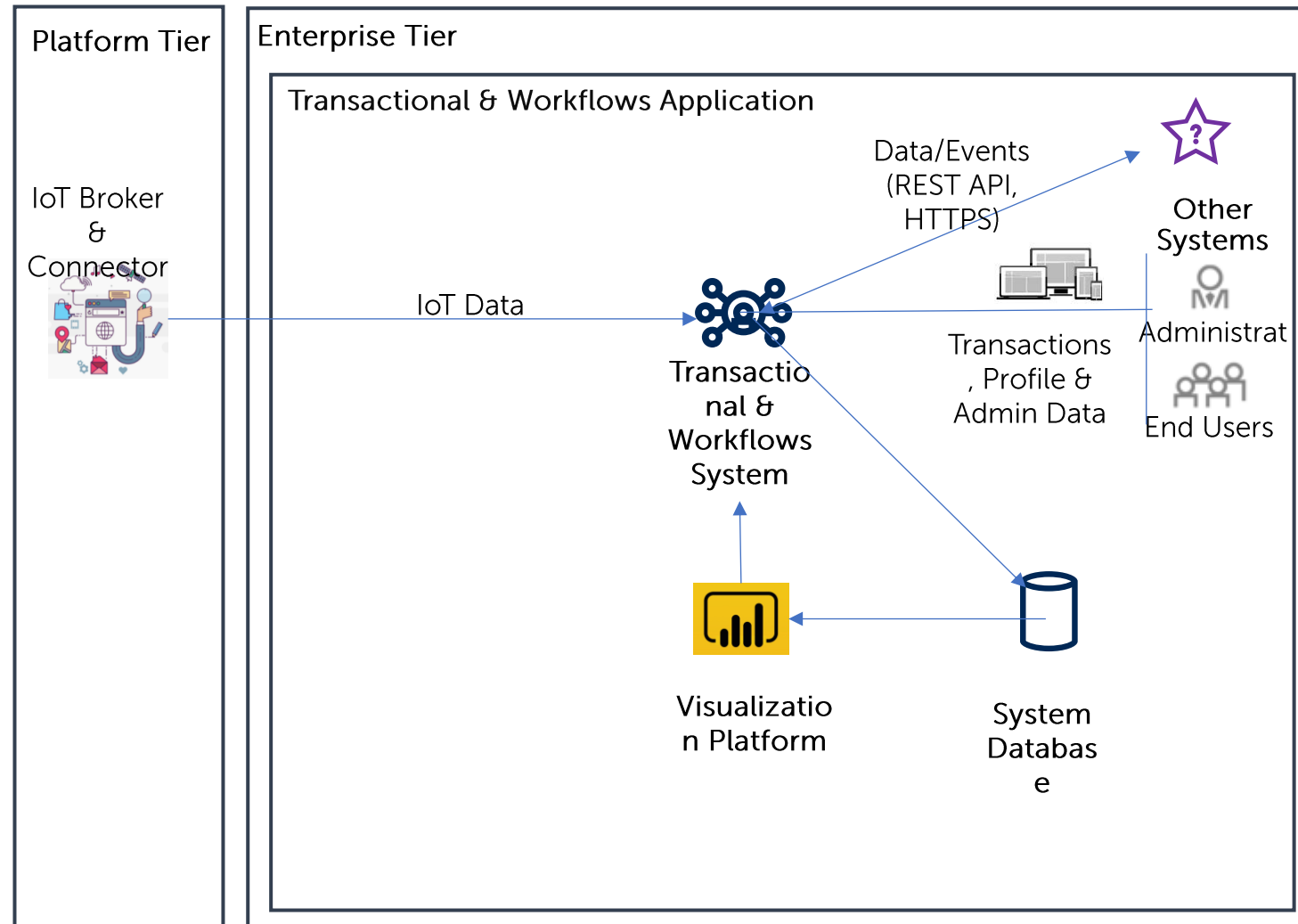
Enterprise Tier

Enterprise tier is where IoT integrates with the set of applications, processes and services required to accomplish a business objective.



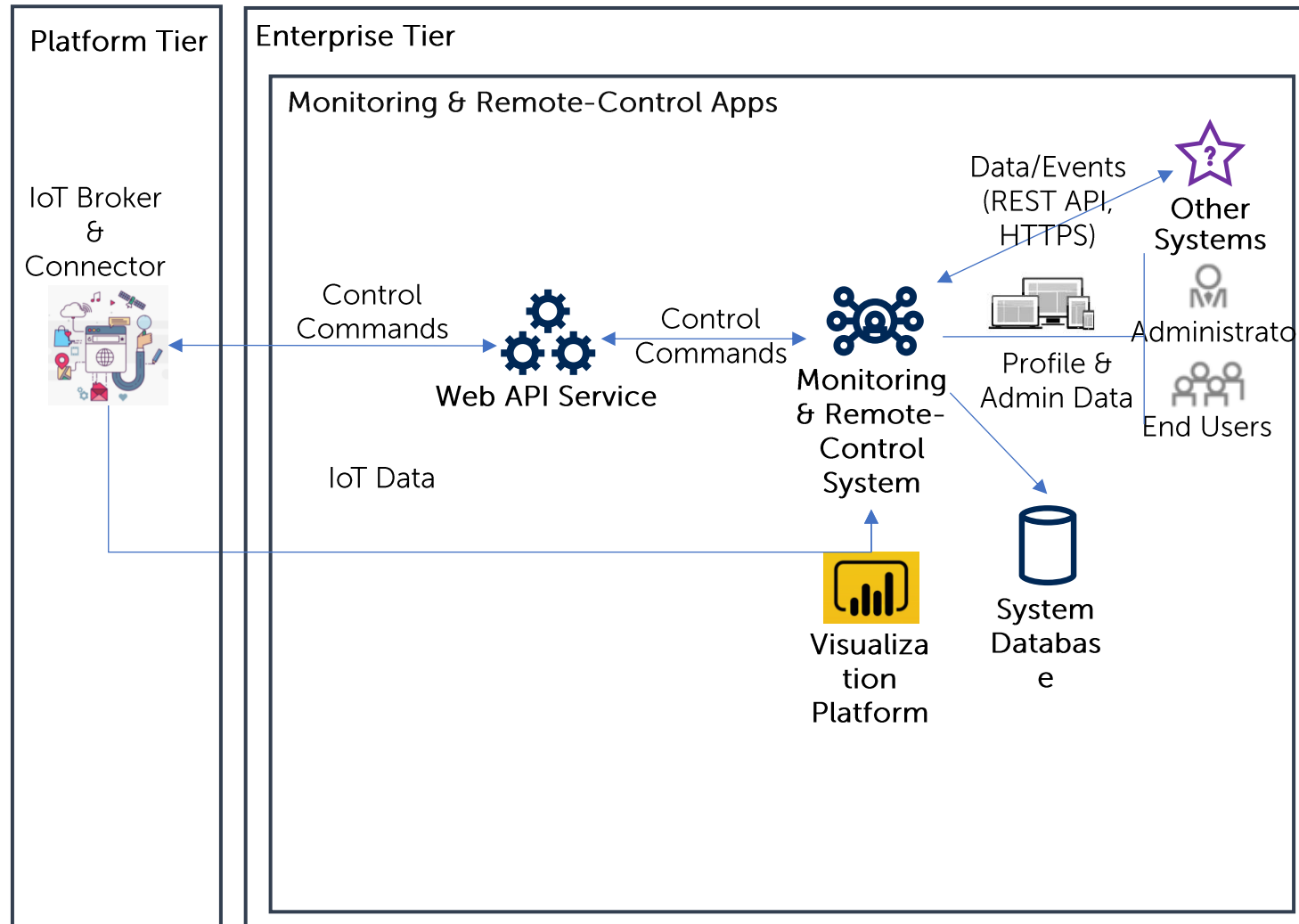
Enterprise Tier

Transactional & Workflows Apps



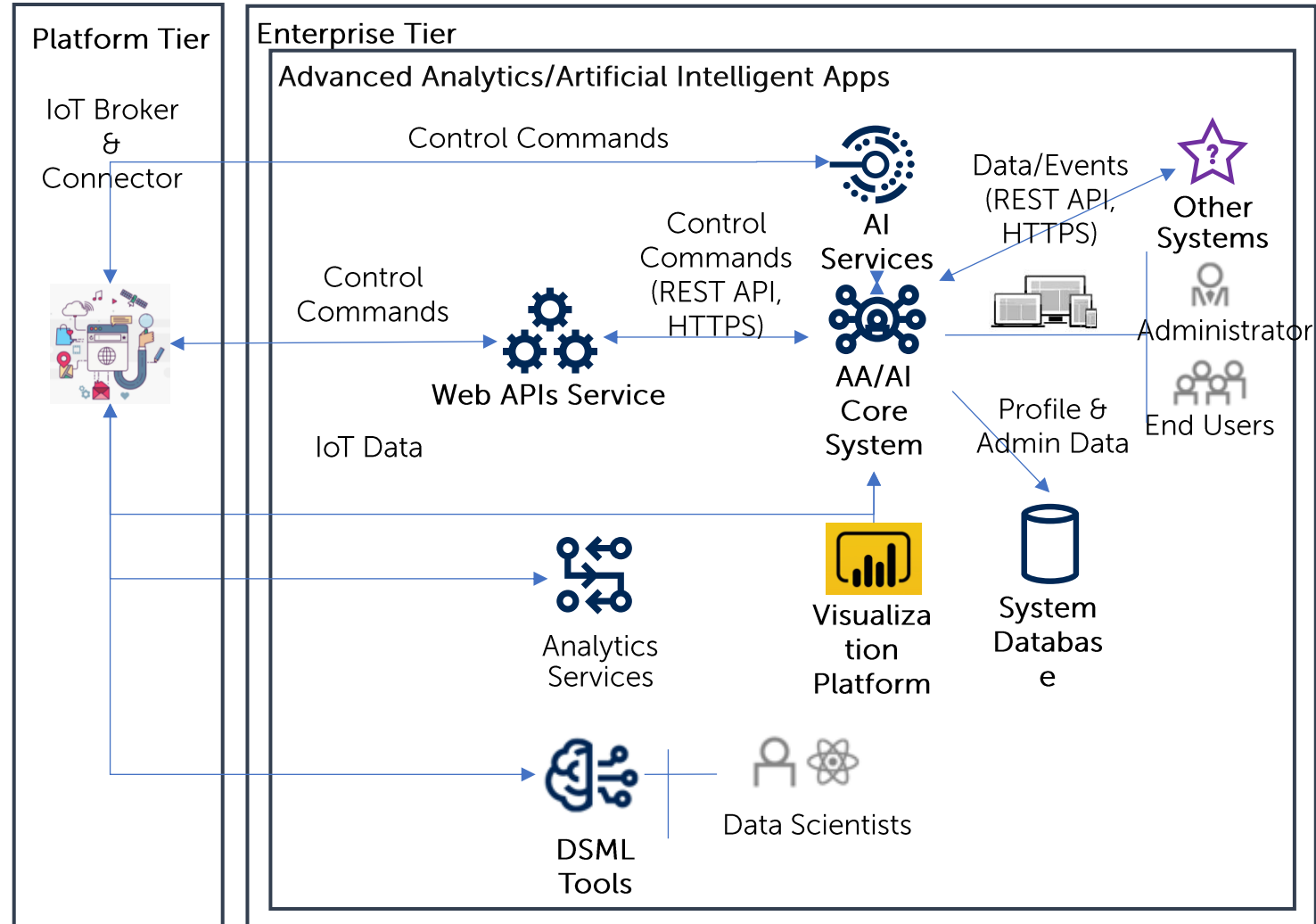
Enterprise Tier

Monitoring & Remote-Control Apps



Enterprise Tier

Advanced Analytics/Artificial Intelligent (AI) Apps



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IoT Cyber Security

1. Risk Assessment on IoT Solution
2. IoT Device Lifecycle Management (Device Onboarding, Provisioning, Configuration, Maintenance, Device Deprovisioning, Logging & Monitoring)
3. Device Security : Device ID & Certs, Device Posture
4. Vulnerability & Patch Management
5. Segmentation : IoT endpoint network segmentation to isolate IoT ecosystem
6. Identity and Access Control on IoT Devices
7. Data Security - Encryption for Data At Rest & In Transit
8. Secure Communications & Integrations (API)
9. OT Interfacing (PTG - Secure IIoT)

